

Enhancing data reliability

DATA MANAGEMENT CONCEPTS



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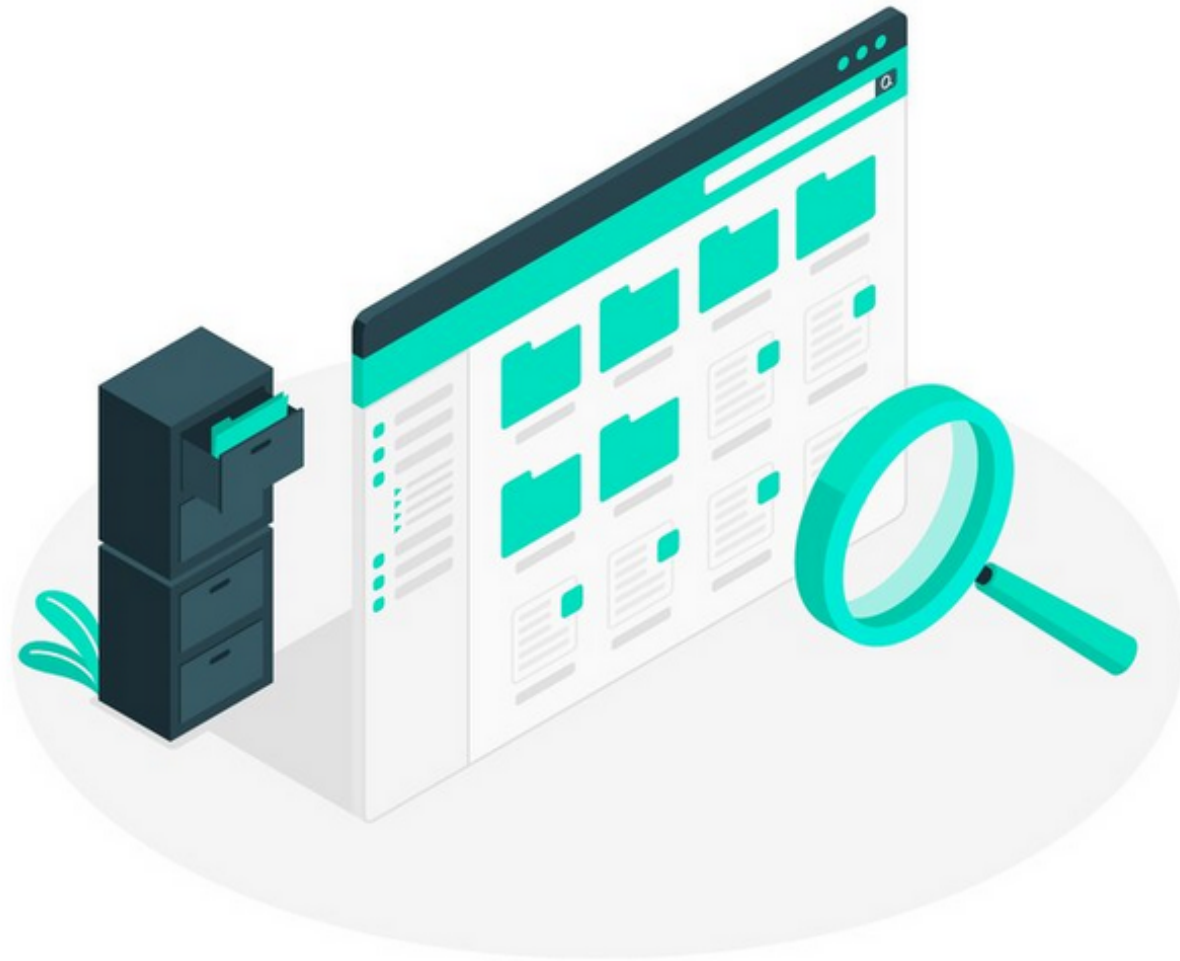
Basics of master data management

- Master data is shared across an organization, such as customer data, product data, employee data, and supplier data
- Ensures the **consistency, accuracy, and coherence** of master data
- Activities include:
 - Data profiling
 - Data consolidation
 - Data validation



¹ Images in this lesson were created using Freepik

Navigating reference data



- **Reference data** is non-transactional and used to categorize other data
- Includes currency codes, country lists, or product categories
- **Reference data management activities** include:
 - Defining data
 - Categorizing data
 - Maintenance and updates

Governance of data about data



- **Metadata** describes various aspects of data, such as its origin, structure, relationships, and usage
- Activities and tools include **schemas, data dictionaries** and **maintenance processes**
- Improves data **understanding** and data **discovery**

Achieving information precision



- **Quality data** is free from errors, duplicates and inconsistencies
- **Data quality management process:**
 - Data profiling
 - Data cleansing
 - Data validation
 - Monitoring and governance

Let's practice!
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Protecting, guiding, and harnessing data wisdom

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Keeping data secure



- **Data security** - protection of data from unauthorized access, corruption, or theft
- Security measures include: **encryption, access controls, firewalls, anti-virus software**
- **Data privacy** - protection of personal information to prevent misuse or unauthorized access
- Privacy measures include: **data anonymization, consent management, privacy policies**

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Elements of data governance

- Data governance establishes a formal framework for decision-making, control, and accountability around data assets
- Involves creating data **policies**, assigning **stewardship** responsibilities, and ensuring regulatory **compliance**
- Frameworks like DAMA-DMBoK or EDM Council DCAM offer a structured approach
- Data governance is a **subset** of data management



Unlocking insights

- **Business intelligence** provides historical, current, and predictive views
- Utilizes components such as **data warehousing, reporting, and dashboards**
- **Data science** extracts insights from both structured and unstructured data
- Employs **machine learning, big data analytics, computer science, and predictive modeling**



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The foundational pillars

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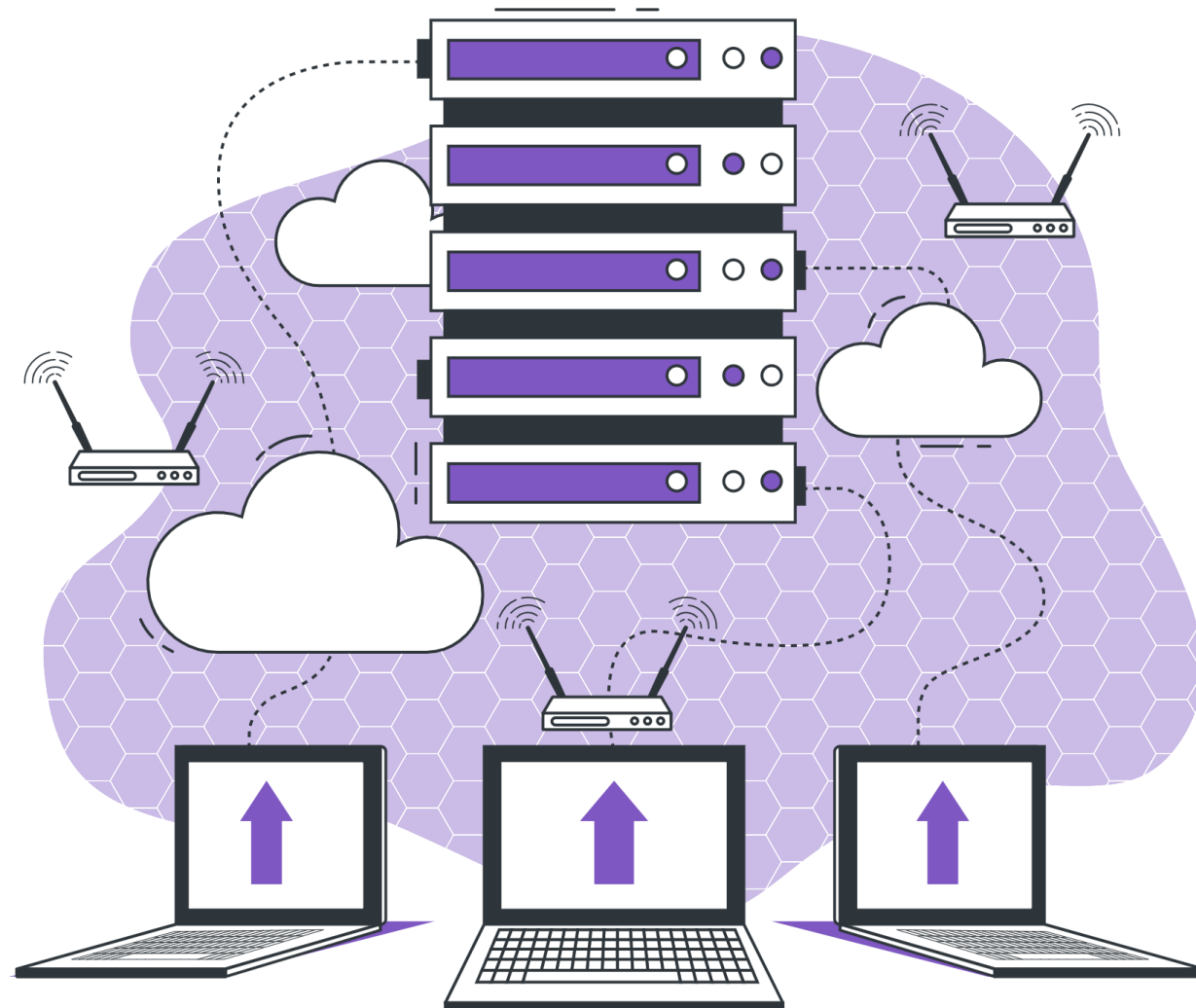
Architecting excellence in system design

- Data architecture includes models, policies, rules and standards for data assets
- Frameworks guide development and implementation e.g. **TOGAF**
- Scalability
- Adaptability
- Flexibility
- Adherence to policies
- Effective metadata management
- Robust security



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Unified information flow



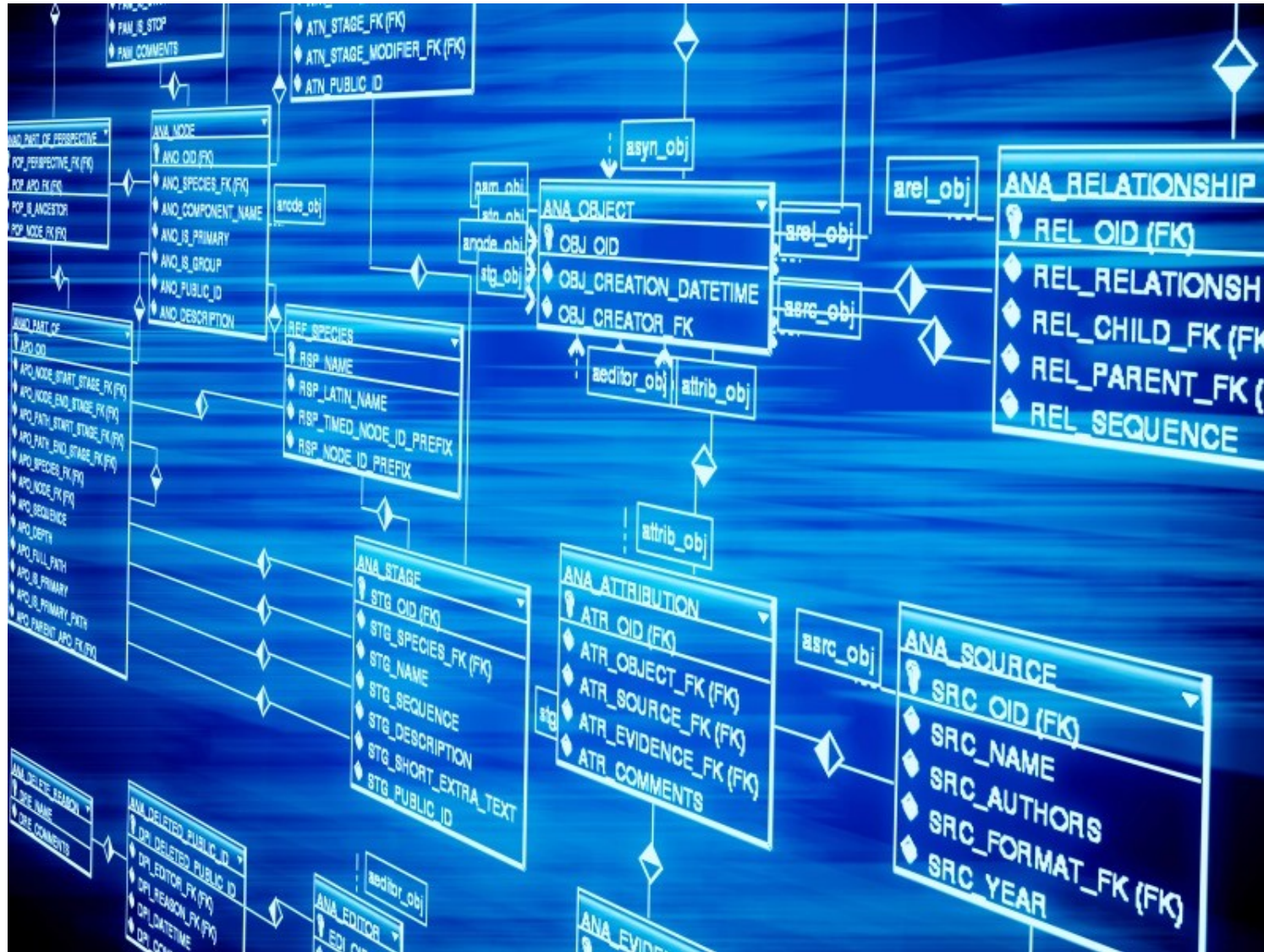
- Data integration unifies diverse data sources
- This reduces **redundancy, complexity** and **destroys silos**
- Activities include **ETL processes, data replication, real-time integration**

Real-time data integration



- Real-time integration uses **streaming data sources, middleware solutions, and specialized integration tools**
- Important to track **data quality, latency, volume and system uptime**
- Data retrieval time should be **minimal**

Modeling with precision



- Involves definition of data requirements and creation of data models
- Data model types: **conceptual, logical, physical**
- Models usually have **entities and attributes**
- **Data modeling** is iterative evolving with requirements

Elevating data storage

- Database and storage management includes organization, optimization and safeguarding of data repositories
- Involves implementing DBMS
- Activities include **effective indexing**, **data compression**, and **archiving** strategies



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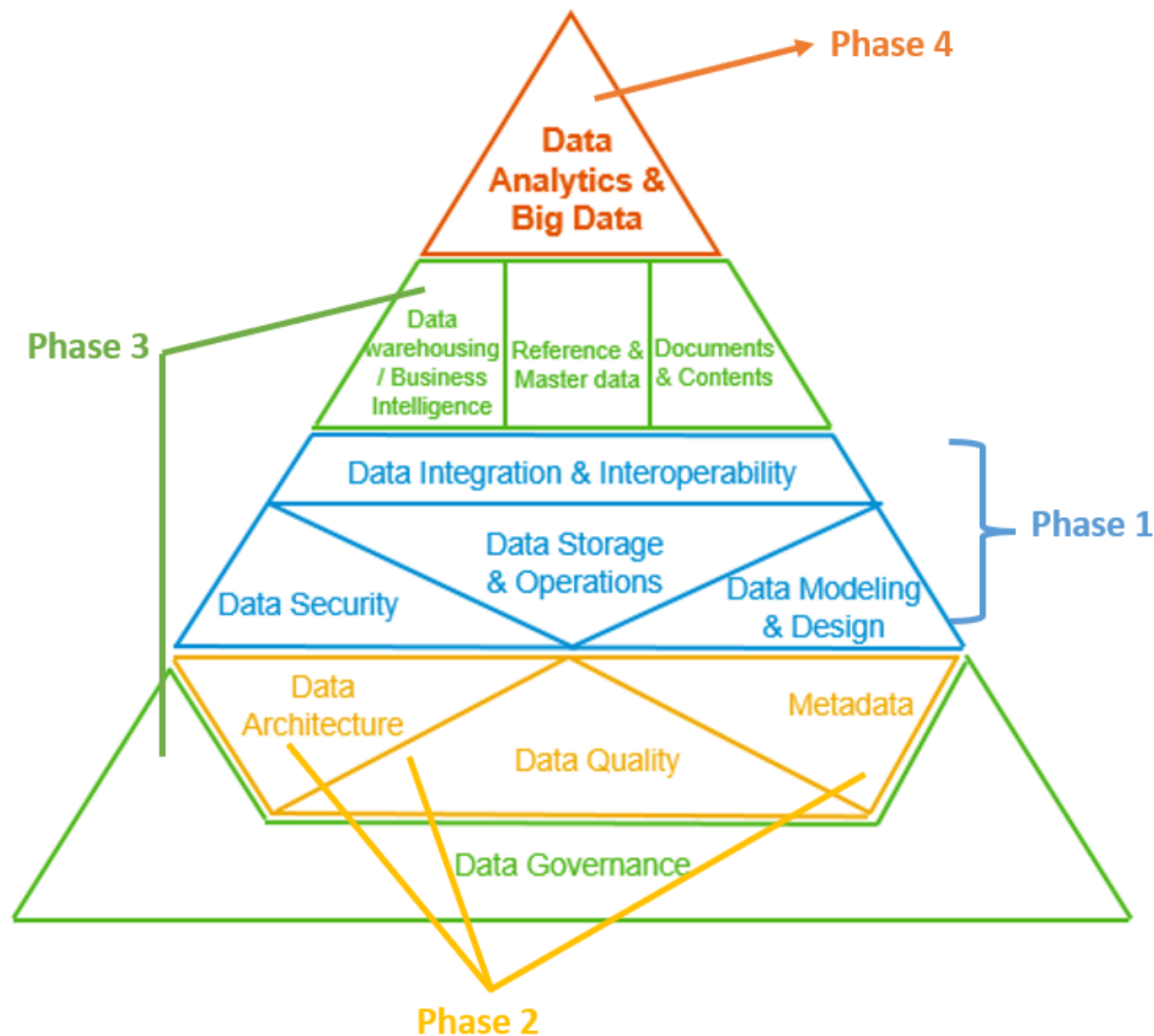
Aiken pyramid

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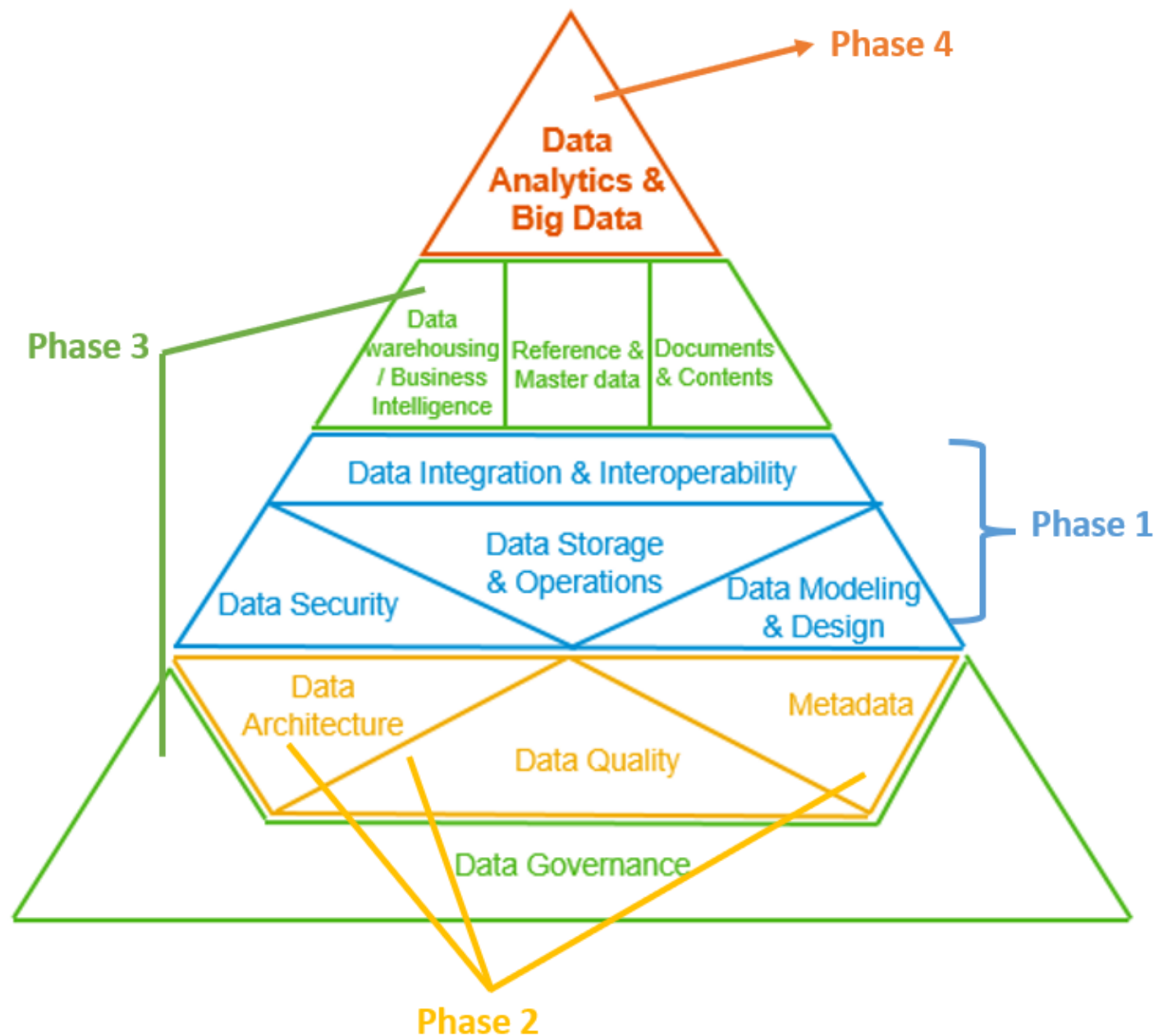
What is the Aiken pyramid?



- Peter Aiken crafted the **Aiken Pyramid** drawing from DMBok functional areas
- Organizes the data management pillars to help organizations understand and manage their data
- Acts as a guide to formulate a **data strategy**
- Entails **four distinct phases**

¹ International, D. D. (2017, July 1). Dama-Dmbok: Data Management Body of Knowledge.

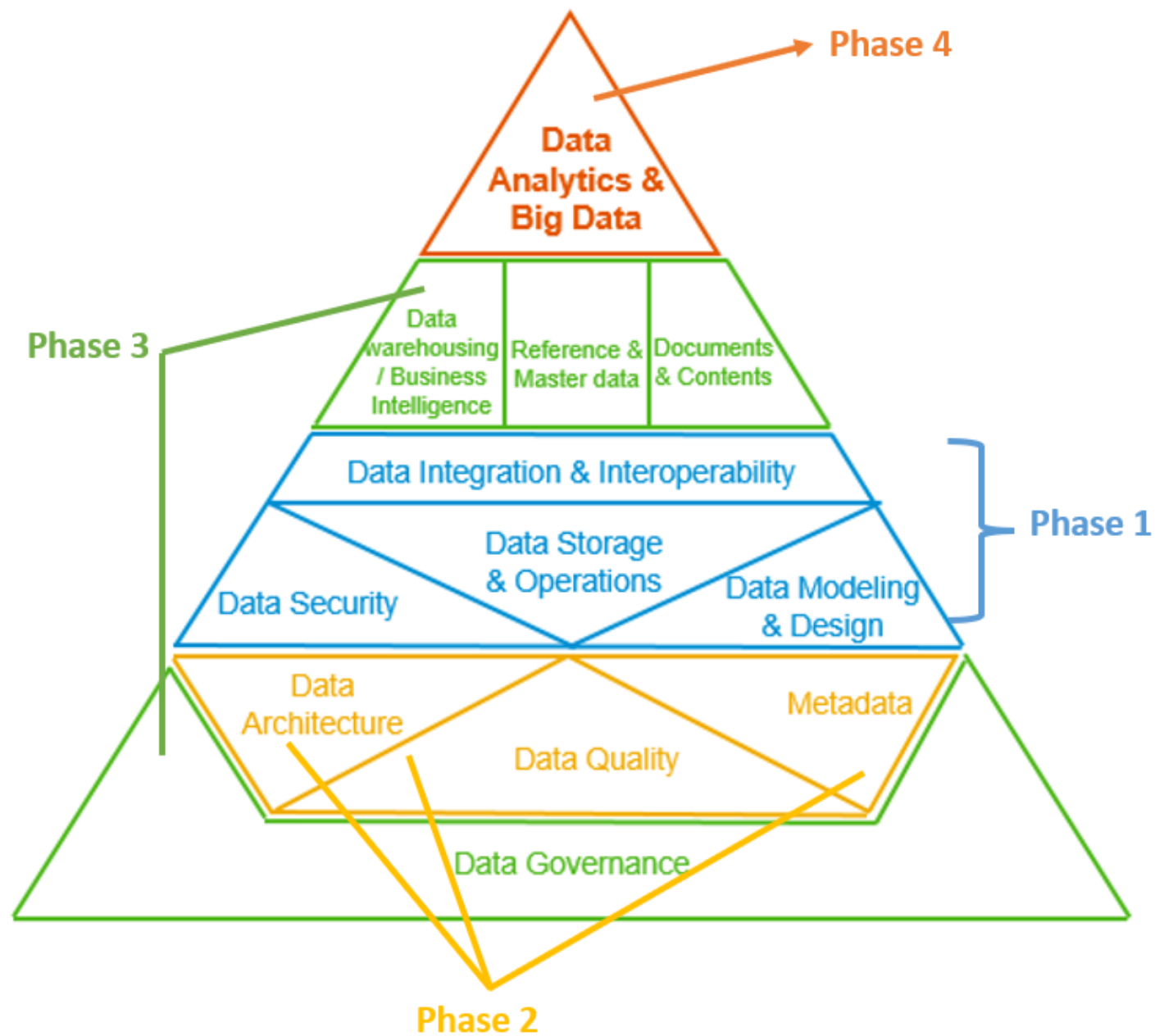
Laying the foundation



- Phase 1 entails:
 - Data integration
 - Data modeling
 - Database and storage management
 - Data security and privacy
- Organizations customize their approaches and solutions to fit their needs
- Sets the **foundation** for later efforts

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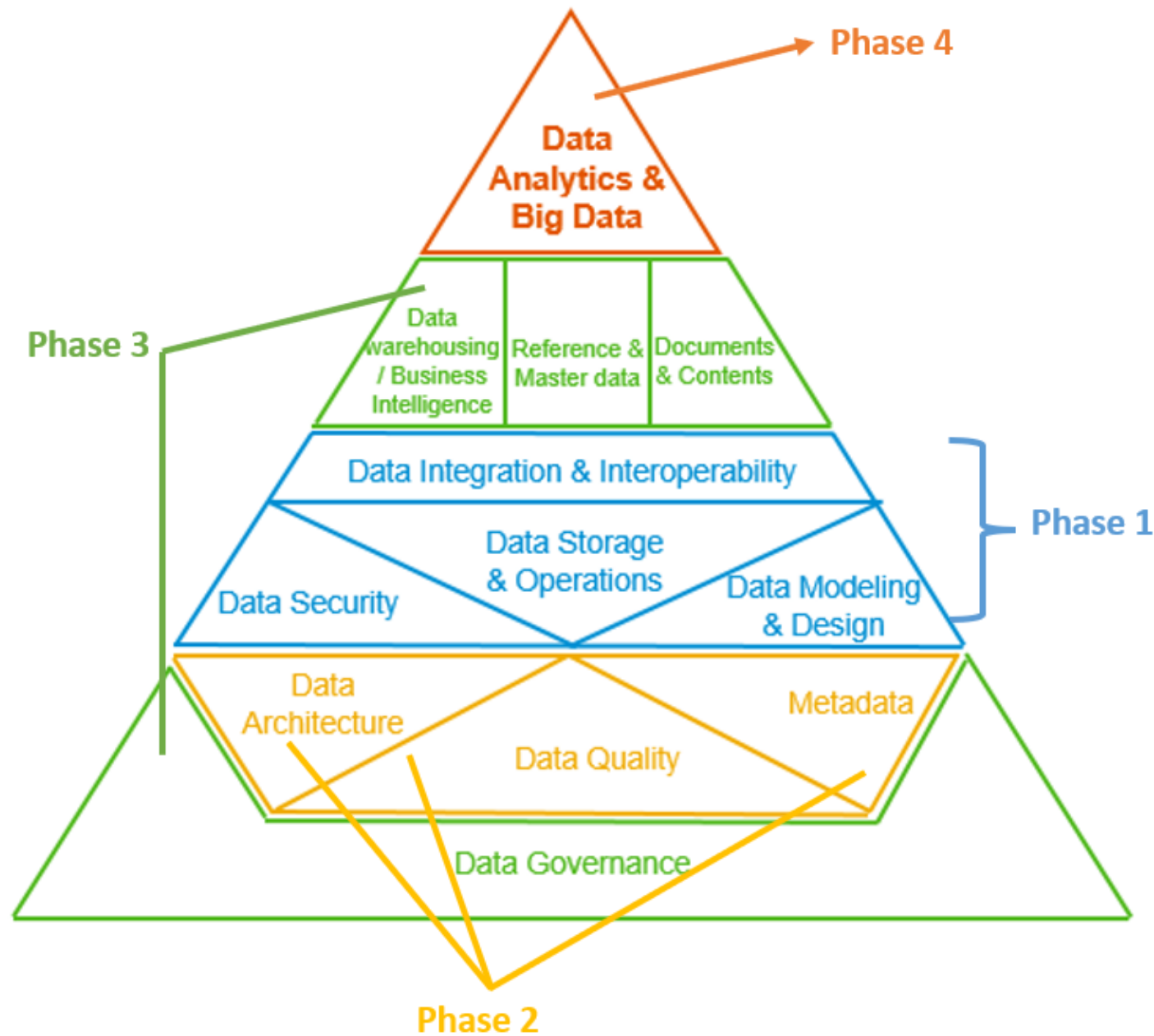
Building the structure



- Phase 2 focuses on:
 - Data architecture
 - Metadata management
 - Data quality
- Aims for **dependability** and **transparency**

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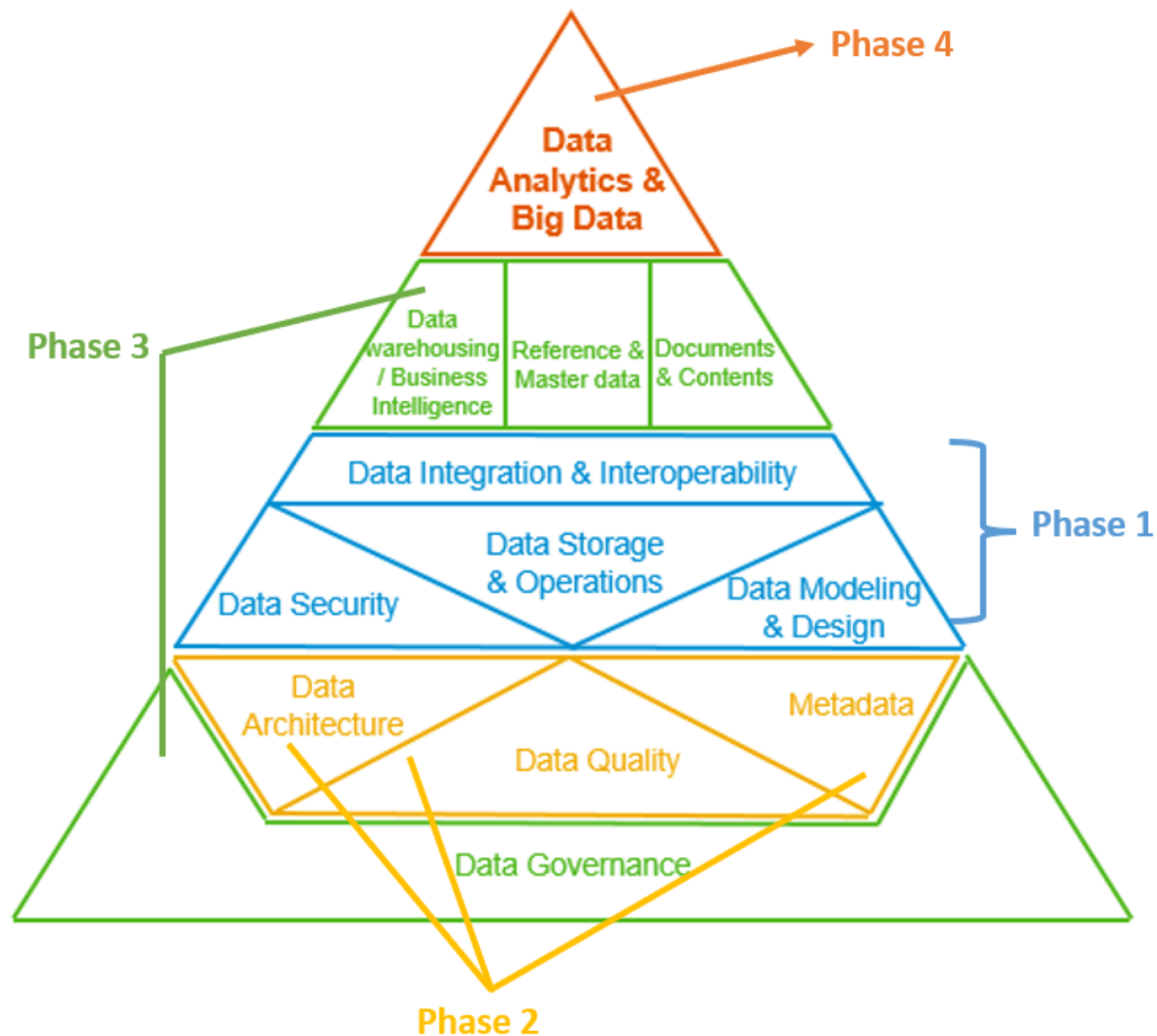
Ascending the pyramid



- Phase 3 requires the incorporation of **data governance** to regulate the previous phase's focus areas
- Governance also empowers:
 - **Reference data management**
 - **Master data management**
 - **Business intelligence**

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Submitting excellence



- At the apex, is **phase 4** centering on **data science**
- Valuable insights support **intelligent business decisions**
- Provides an **understanding of the data landscape**

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Flexibility



- Organizations can tailor their own data management approach
- Influenced by:
 - **Industry requirements**
 - **Technological capacities**
 - **Budget constraints**
 - **Immediate business objectives**
 - **Unforeseen challenges**
- Approaches can be **strategic**

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